

Logical Structured Context Worlds: Exact Context Boundaries Without a Runtime

Declarative SCW boundaries over Maxey0's five-tier memory, with user-side ChatGPT advertising privacy as a first application, tested on a real ChatGPT data export

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human@maxey0.com · 30 September 2026 · SCW0 project: SCW Privacy · research paper with a working application (version 3)

Abstract

A Maxey0 Structured Context World (SCW) is a bounded, addressable context with explicit rules for what may enter and leave it. Where the owner of a boundary does not control the runtime (inside another vendor's product, between tenants sharing a context store, or between agents sharing one model context), the boundary cannot be a process or a sandbox. This paper defines the logical SCW: an SCW whose boundary is a declarative policy object, enforced by one pure evaluator on every egress path, dominated by a fixed set of invariants that no configurable field can weaken, and required to name the rule behind every withheld signal. We pair the pattern with Maxey0's memory model of five typed tiers (Working, Episodic, Semantic, Persistent, Procedural) joined by an evidence-gated promotion gate, so that whole kinds of memory can be removed from an egress path by construction rather than by a setting. As a first application we build a user-side advertising boundary for ChatGPT Free and Go users, whose only ad controls today are global switches [1]. The application, Maxey0 SCWs for ChatGPT Ads (version 3), is one self-contained React file with no server, no storage and no user data. A seeded property test of 20,000 random profiles, worlds, messages and histories found zero violations of 19 per-case checks, and each of seven deliberately injected faults was caught. Replaying one real export of 602 chats (9,295 user messages) through the evaluator on its owner's computer produced zero invariant breaches under all three presets and suppressed 40 chats that touched a sensitive topic. A logical SCW is exactly as strong as the guarantee that all egress passes through its evaluator; in the ChatGPT application it cannot bind OpenAI's ad system, so it gives the user an exact, inspectable account of their own policy, points to the controls OpenAI does enforce, and labels everything else as advisory.

Keywords: Structured Context World, SCW, logical SCW, logical isolation, declarative boundaries, multi-agent context, conversational advertising, ChatGPT ads, privacy controls, typed memory, promotion gate, data minimization, information-flow policy, user-side privacy tooling, Maxey0

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1 Introduction: advertising arrives in the context window

An assistant that carries advertising reads the same context the user writes into. OpenAI's ad selection uses the current chat and basic context such as general location and language; when ad personalization is on, it may also use how the user interacts with ads, and past chats and memory [1].

The user-facing controls are coarse. A person who is happy for shopping questions to shape their ads may not want a work conversation, a study session or a question about a medication to do the same. What such a person needs is a boundary per kind of conversation, with a precise statement of what crosses it. That is the job the Maxey0 architecture gives a Structured Context World [5]: an explicit, addressable context with rules for what may enter and leave.

A user cannot place a process boundary inside someone else's product. They can, however, state a policy exactly, compute its consequences on their own data, and carry the result to the switches that do bind the provider. This paper calls an SCW built that way a logical SCW. Its boundary is an object, not a runtime; its enforcement is a function, not a sandbox.

The claim of this paper

A context boundary can be made exact without being physical, provided that (a) it is written as a declarative object, (b) one pure evaluator decides every egress, (c) a small set of invariants dominates every configurable field, and (d) the evaluator explains every withheld signal. The boundary is then exactly as strong as the guarantee that all egress passes through the evaluator, and no stronger.

This revision makes three changes to earlier drafts. It replaces a four-level memory stack with Maxey0's current memory model of five typed tiers and a promotion gate (Section 4). It rebuilds the application around that model, with visualizations that let a non-specialist see the boundary (Section 6). And it re-measures everything with the new evaluator, including a fresh replay of one real ChatGPT export and a new property test (Section 7).

2 Background and research position

2.1 Structured Context Worlds and SCW0

In the Maxey0 execution-space model, an SCW is a bounded semantic and contextual space that binds context, identity, capabilities, resources, provenance, admission rules and permitted transitions into one addressable unit; SCW0 is the supervisory layer that creates SCWs, routes context between them and evaluates boundary transitions [5]. The System One paper separates decision from authorization, execution and observation [6]. The logical SCW in this paper is a small, concrete instance of both ideas: the policy object is the authorization boundary, and the withheld-signal ledger is the observation record.

2.2 What OpenAI documents

OpenAI's Help Center states that ads may appear on the Free and Go plans and not on Plus, Pro, Business, Enterprise or Edu; that accounts identified as belonging to users under 18 do not see ads; that advertisers receive only aggregated, non-identifying performance information such as views or clicks; that ads are not eligible to appear near sensitive or regulated topics, including personal health, mental health or politics; that users can turn off ad personalization, stop past chats and memory from informing ads, see why an ad was shown and clear ads data; and that Temporary Chats do not show ads [1]. A Temporary Chat does not create or update memories and stays out of chat history [2]. The data export is requested from Settings, Data controls, and arrives as a zip file [3].

If advertisers see only aggregates, the privacy-relevant boundary is not between the user and the advertiser. It is between the user's context and the ad-selection system that reads it. The application therefore models what the selector may read, and the evaluation reports what would cross each world toward it.

2.3 Earlier versions

The first version of the application, built by the Maxey1 agent, set out the user-facing idea: privacy gates for kinds of conversation, each with ad limits, category allowances, data-sharing switches and a list of accessible memory tiers, plus a JSON profile to paste into ChatGPT. It had three problems. Its tiers (Scratchpad, Episodic, Persistent, Permanent) were a duration stack rather than kinds of memory. Its privacy score was a sum of fixed points for switches. And it claimed an enforcement it could not deliver, for example that "ChatGPT advertisers must respect your SCW containers." Versions 2 to 2.3 introduced the evaluator, invariants and an in-page export audit, but kept a four-level tier ceiling (L1 to L4). Version 3, described here, adopts the five typed tiers, keeps the profile format backward compatible, and states on every relevant screen which channel is enforced and which is advisory.

3 The logical SCW

A logical SCW is a tuple of declarative fields. A separate, global set of invariants applies to every SCW, and no SCW field can weaken it.

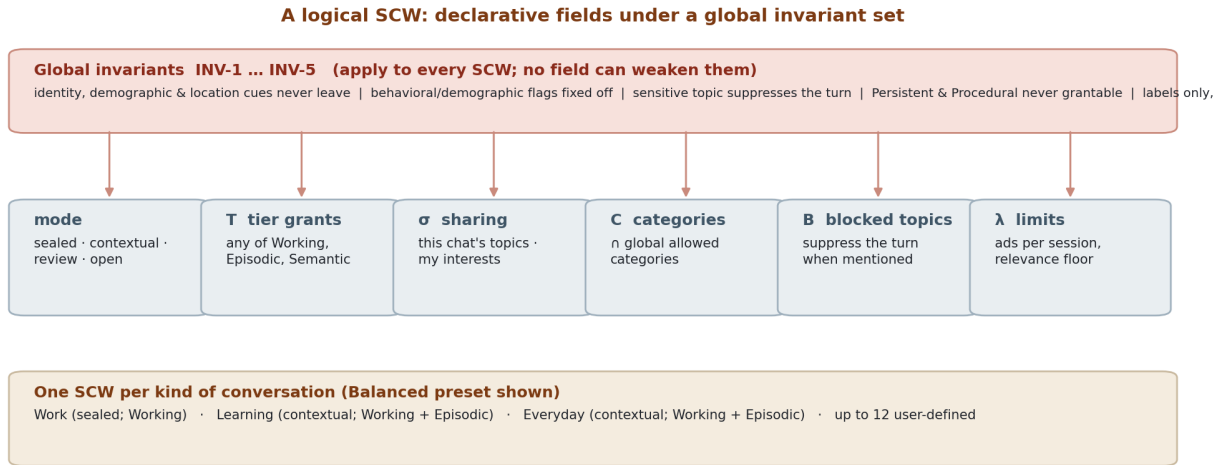


Figure 1. The logical SCW object: six configurable fields per world, dominated by one global invariant set.

$$\text{SCW} = (\text{id}, \text{mode}, \text{T}, \sigma, \text{C}, \text{B}, \lambda) \text{ under } \text{INV} = \{ \text{INV-1} \dots \text{INV-5} \}$$

Field	Meaning	Values in the application
mode	How the world treats egress	sealed (nothing leaves); contextual (topic labels only); review (a matched ad is held for approval); open
T tier grants	Which memory tiers the selector may read in this world	any subset of Working, Episodic, Semantic. Persistent and Procedural cannot be granted.
σ sharing	Which sources may be used	this chat's topics (Working, Episodic); stated interests (Semantic); each also gated by a profile-wide switch
C, B	Allowed categories; blocked topics	four ad categories, intersected with the profile's; free-text topics; single words matched as whole words, phrases as substrings
λ limits	Per-world ad count and relevance floor	combined with the profile-wide hourly limit and relevance floor; the higher floor applies
Invariant	Statement	
INV-1	Identity cues (email, phone, name) never leave any SCW. Age, occupation or income, and location cues are withheld under the same rule.	
INV-2	Demographic and behavioral data never leave any SCW; the corresponding flags are fixed to off and cannot be set by editing or import.	
INV-3	A sensitive topic (health, mental health, politics, religion, sexuality, financial hardship) suppresses every release and every ad for that turn.	
INV-4	Persistent and Procedural memory can never be granted to an SCW; grant lists are normalized on every edit and every import.	
INV-5	The selector receives vocabulary labels and categories only: never message text, identity or history.	

Tier grants are a set rather than a ceiling on purpose. The tiers are kinds of memory, not levels of one quantity, so there is no order in which reading Semantic should imply reading Episodic. Instead, exposure is made explicit by weights (Working 0.2, Episodic 0.3, Semantic 0.5) that the

application shows alongside every world, and the two tiers whose contents identify or instruct the person are removed from the grantable set altogether.

3.1 Signals, not text

The evaluator never releases text. Each input is first reduced by a deterministic keyword classifier to labels from a fixed vocabulary (such as `kubernetes` or `trail`), each with an ad category, plus an optional sensitive class and a list of identity-cue kinds. A released signal is a label, its category and the tier it came from. The most the selector could ever receive is a vocabulary label, or an interest the user typed (at most 40 characters and four words, enforced on entry, on import and again by the evaluator) that the classifier could place in an allowed category.

3.2 The evaluator and its rule order

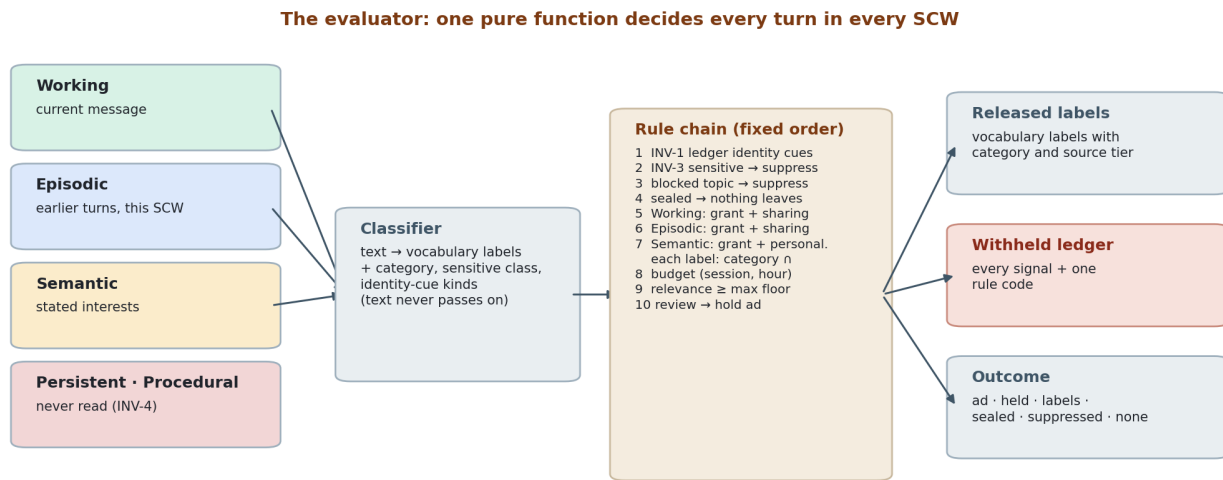


Figure 2. The evaluator. Three grantable tiers feed one fixed rule chain; the sealed tiers are never read. It returns released labels, a ledger of withheld signals with one rule code each, and an outcome.

The evaluator takes the profile, one SCW, the current message (Working), the earlier messages in the same SCW's session (Episodic), the ads already shown and the profile's stated interests (Semantic). Rules run in a fixed order and the first that ends the turn is reported. Identity cues are always written to the ledger first. A sensitive topic, or a blocked topic, then suppresses the turn; a sealed world releases nothing. Otherwise each tier is read only if it is granted and its sharing switches are on, and each label is released only if its category is allowed by both the world and the profile. Earlier turns that were themselves sensitive contribute nothing to Episodic. Ads are matched only against released labels, from a fictional inventory, and one is served only if the brand is not blocked, the session and hourly budgets are not spent, and its relevance (55 + 20 per matched label, +10 if a match came from the current message, capped at 100) reaches the higher of the two floors. In review mode a matched ad is held for the user.

3.3 Properties

Property	Statement	How it is checked
Determinism	The same inputs always yield the same decision.	Property test (re-evaluation per case)
Invariant dominance	No released signal is from a sealed tier, identity-shaped or sensitive; a sensitive turn releases nothing; no setting of any field changes this.	Property test, 20,000 cases; real-export replay
Grant and sharing soundness	A label is released only from a granted tier whose sharing switches are on, and only in an allowed category.	Property test
Minimization	Released labels are vocabulary items or bounded user interests: no label longer than 40 characters or four words.	Property test, with an over-long interest in the input pool
Mode containment	Sealed releases nothing; review never serves an unapproved ad.	Property test
Ad soundness	A served ad respects category, relevance floor, budget and blocked brands.	Property test
Explainability	Every identity cue is ledgered as INV-1; every withheld signal carries one rule code.	Property test; visible in the simulator

4 Memory: five typed tiers and a promotion gate

Maxey0 memory is not a duration stack. It has five tiers typed by what they hold, and items move between tiers only through an evidence-gated promotion gate.

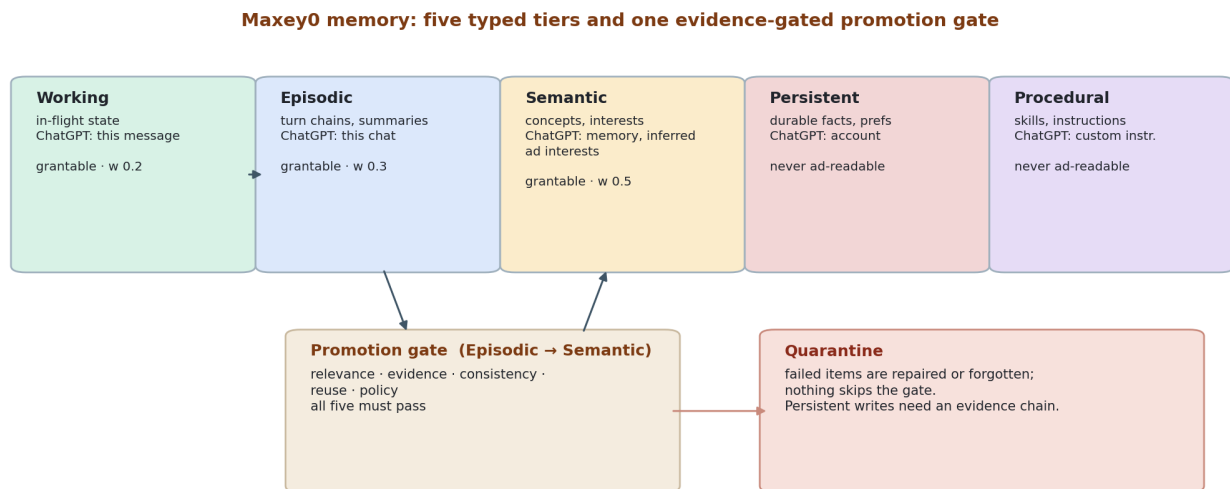


Figure 3. The five tiers, each with its ChatGPT counterpart and its ad-exposure weight, and the promotion gate with quarantine.

Tier	Holds	Lifecycle	ChatGPT counterpart	Ad-readable
Working	volatile in-flight state	TTL-enforced; purged at session close	the message being written	if granted and shared
Episodic	turn chains, session summaries	time-stamped; summarized at close	earlier turns in this chat	if granted and shared
Semantic	concept graph: concepts, interests	admitted only through the gate	saved memory, chat history, inferred ad interests	if granted and personalization on
Persistent	durable facts, versioned preferences	written rarely, with an evidence chain	account identity, profile, settings	never
Procedural	skills and instructions	creation, revision, deprecation	custom instructions; this profile	never

The promotion gate checks five things before an item enters Semantic memory: relevance, evidence, consistency, reuse and policy. Items that fail go to quarantine, where they are repaired or forgotten; nothing skips the gate. The model matters for advertising privacy because the Semantic tier is where consolidated facts about a person live, and it is the tier that ChatGPT’s “past chats and memory” setting governs. Section 5 shows that a real export also contains a third Semantic object: the interests the ad system itself inferred.

In the application, the Semantic tier is represented by the user’s stated interests. The promotion gate is illustrated, not executed: the application does not promote items between tiers on its own, and every interest is entered, and can be removed, by the user.

5 What a ChatGPT export holds, by tier

The first author requested a ChatGPT data export (the folder is dated 11 September 2026) and supplied it as a dataset, on the condition that none of it be built into the application.

To learn what an export contains we read its schema (key names, value types, string lengths) and computed aggregate counts on the owner’s computer, during the work on version 2 (29 September 2026). No message text, interest text, email, birth year or account ID was displayed to the authors’ tools or copied off the machine. The folder held 972 files in 1.5 GB; ten of them carry what this problem needs (Appendix A).

File	Content (schema)	Tier
user.json	email, birth_year, chatgpt_plus_user, id	Persistent
user_settings.json	settings incl. training_allowed	Persistent
ads.json: ads_profile	inferred_profile.user_interests[] (4 items), profile_version	Semantic
ads.json: ad_history, ad_hides, reported_ads	event_name, event_time, advertiser_name, ad_title	Episodic
conversations-000...006.json	messages; memory_scope; is_do_not_remember	Episodic
saved memories	no file present	—

Three facts about this one export shaped the design. First, it contains the ad system’s own inferred interest profile: Semantic memory the user did not write and may never have seen. OpenAI states that deleted ads data is no longer used to show ads [1]; it does not say explicitly that this list is part of it. Second, it contains no saved-memory file, and none of its 33,673 messages is addressed to or from the memory tool. Third, every chat carries a memory flag: all 602 had `memory_scope = global_enabled` and none was marked `do-not-remember`. The account-level files map onto Persistent memory, which is exactly the tier INV-4 removes from every advertising world.

6 The application: Maxey0 SCWs for ChatGPT Ads

The application is one React file that imports only React and an icon library, carries its own styles, makes no network requests and stores nothing. Every advertiser in it is fictional.

Page	What the user does	What it demonstrates
Overview	Reads the privacy score; picks a preset; follows the exposure flow; opens any replay cell	The whole boundary at once: tiers → SCWs → advertiser or withheld, and 18 test prompts × every SCW
Latent Map	Watches fictional advertisers probe the boundary; clicks a world to edit it	The boundary as a live system: sealed core, tier rings, SCW worlds, admitted and deflected probes
SCW Studio	Edits each world’s mode, tiers, categories, sharing, limits and blocked topics	The control plane, with a tier × SCW grant grid and a live preview per world
Simulator	Chats inside a chosen SCW	Each rule with pass or fail, released and withheld signals with rule codes, a fictional ad, and the same message in every SCW
Advertiser Lens	Chooses a world and a prompt	The complete payload the selector would receive; text, identity, history and location are null
Memory	Sees the five tiers and the animated promotion gate; edits interests	The typed memory model applied to the user’s own Semantic tier
Learn SCWs	Reads five short lessons	Logical SCWs explained, each lesson linked to the page that shows it
Apply to ChatGPT	Sets global controls; reviews ChatGPT’s settings; copies the instruction; exports or imports the profile	The split between enforced and advisory channels

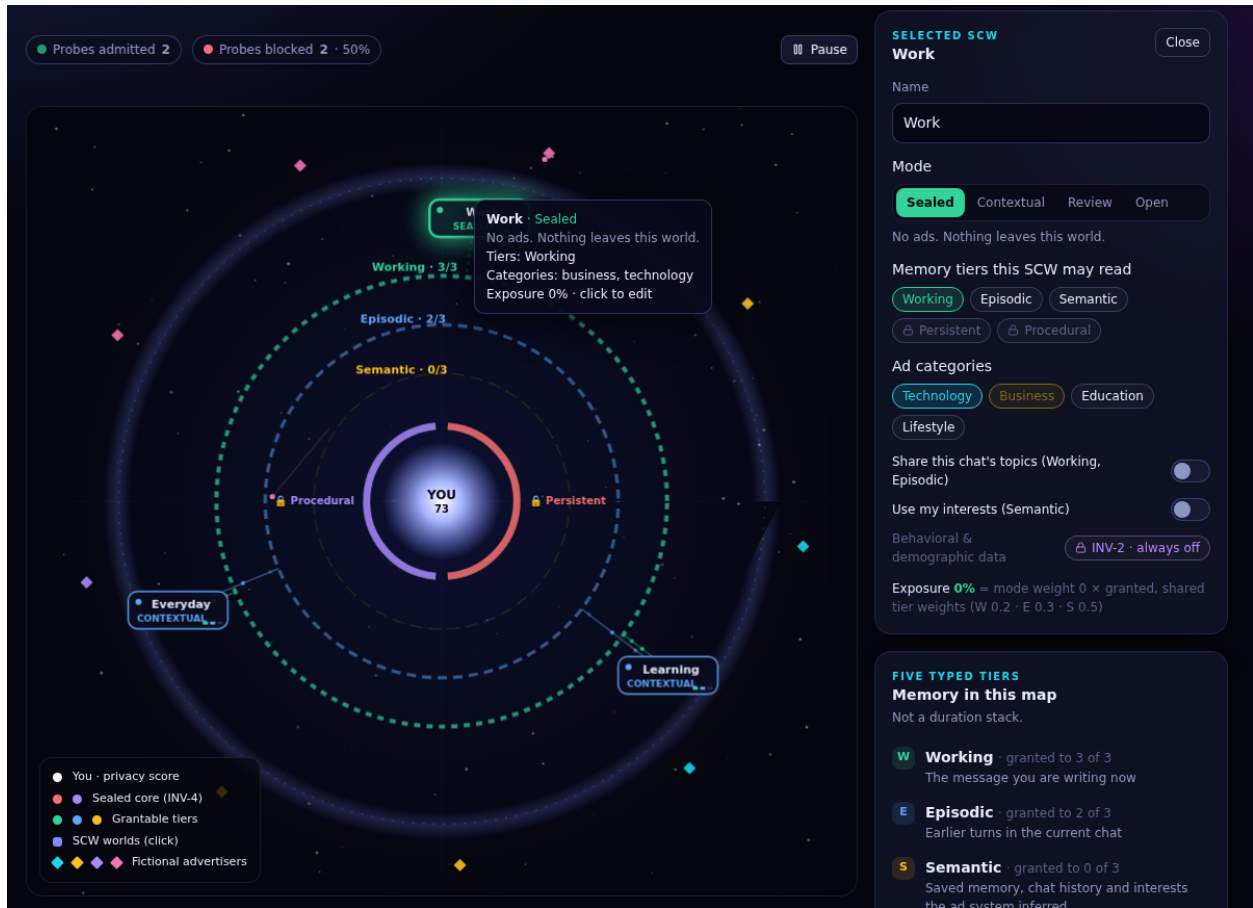


Figure 4. The Latent Map (Balanced preset, one world selected). The sealed core (Persistent, Procedural) surrounds the user; three dashed rings are the grantable tiers, labelled with how many worlds read them; SCW worlds orbit with coloured channels to the tiers they read; fictional advertisers on the rim fire probes that are admitted or deflected by a simplified form of the evaluator’s rules.

6.1 Visualizations that teach the boundary

The application is meant to teach logical SCWs to a general audience, so each visualization is computed from the live policy rather than drawn as an illustration. The exposure flow is a flow diagram from every tier and from identity cues into every world, and from each world to the advertiser or to withheld; band widths count channels, and solid bands are exactly those the evaluator would open. The replay grid runs 18 synthetic prompts through every world and colours each cell by outcome, with an invariant-breach counter that should always read zero. The Latent Map animates a simplified form of the evaluator’s admission rules (mode, categories, the Working grant and its sharing, blocked brands; it ignores Episodic, Semantic, relevance, budget and the review hold), so a change to a world’s mode or categories is visible within a second as a change in which probes pass.

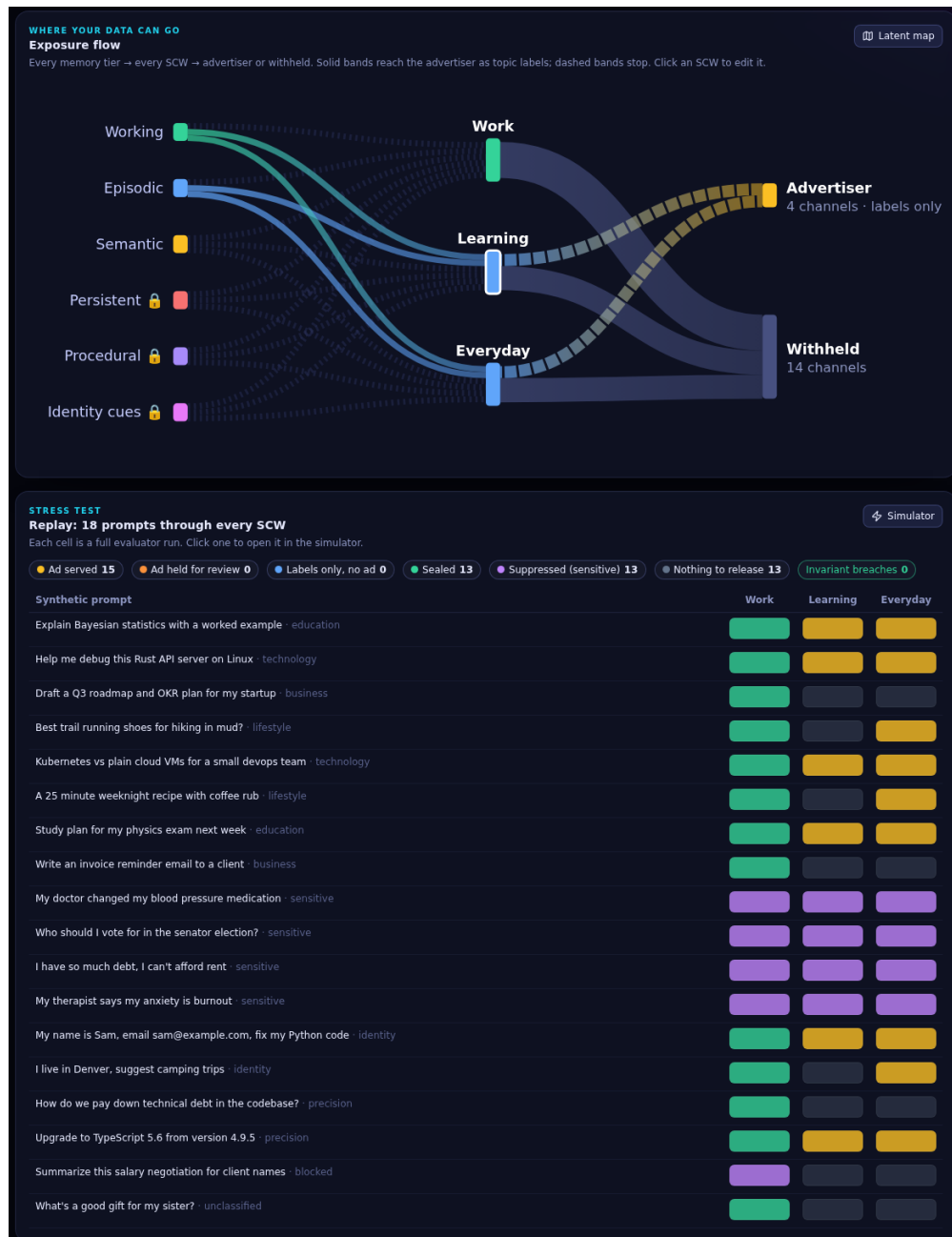


Figure 5. The Overview under the Balanced preset: exposure flow (top) and the replay of 18 synthetic prompts through every SCW (bottom). The breach counter reads 0.

6.2 Three channels, three levels of force

The settings checklist points to OpenAI’s own controls: memory and chat-history reference, ad personalization, clearing ads data, and Temporary Chat for anything that belongs in a sealed world [1][2]. Only these change what OpenAI’s systems read. The instruction block restates the worlds, invariants, categories, blocked brands and limits for the assistant; it can shape what the assistant uses in conversation and cannot change ad selection. The profile file (version 2.0.0) is the user’s portable copy of the policy; import accepts profiles from earlier versions, converts their tier names, drops any grant that would violate INV-4, drops over-long interests, and clamps limits to the ranges the interface allows.

What the tool does not claim

It does not claim that ChatGPT will honor the user's SCWs. It shows exactly what the user's policy would release, points to the switches OpenAI enforces, and labels the pasted instruction as a request.

7 Evaluation

Every number in this section was produced by running code on 29–30 September 2026. The real-export results are a simulation: the user's own chats replayed through the policy on their own machine. They measure the policy, not OpenAI's ad system.

7.1 Tests

Test	Setup	Result
Type check	TypeScript strict mode with unused-locals checking	0 errors
Engine checks	Sensitive suppression, sealed mode, identity withholding, precision cases (technical debt, version strings), tier normalization, Semantic gating, brand blocking, budget, invariant sweep, export-import round trip, rejection of foreign JSON	11 of 11 pass
Browser checks	Headless Chromium at 1440 px and 400 px: every page, map painting and probes, selecting a world from the map, score reacting to grants, undo, simulator trace, lens nulls, legacy import and migration, foreign import rejected	17 of 17 pass; 0 script errors; no horizontal overflow
Property test	20,000 seeded cases (LCG seed 0x5c3a): random preset, world mode, grants (including legacy and sealed tier names), categories, sharing switches, limits, blocked topics and brands; messages and histories mixing technical, sensitive and identity content; interests including sensitive, identity-shaped and over-long strings	0 violations of 19 per-case checks; 4,722 labels released, 220,944 withheld, 5,723 turns suppressed, 246 ads
Mutation check	Seven faults injected one at a time, 5,000 cases each	All caught: drop INV-3 (161 violations); drop sealed mode (128); ignore personalization switches (718); drop the category rule (3,560); allow sealed-tier grants (3,778); serve ads in review mode (30); drop the interest length bound (66)

The property test remains a test, not a proof. The invariants hold for every input by rule order; the test is evidence that the code follows that order. The sealed-tier mutation is caught by the normalization check rather than by a released signal, because the evaluator does not read Persistent or Procedural memory at all; that is a second, independent guard.

7.2 Exposure on the synthetic prompts

The privacy score is $100 \times (1 - \text{mean world exposure})$, where a world's exposure is its mode weight (sealed 0, review 0.6, contextual 0.8, open 1) times the sum of the weights of the tiers it is granted and allowed to share, and zero if the world and the profile share no allowed category. It is a transparent summary, not a standard metric. The 18 synthetic prompts cover the four ad categories, the sensitive classes, identity cues, two precision traps and one blocked topic.

Preset	Score	Released labels	Withheld signals	Outcomes over 54 turns (18 prompts × 3 SCWs)
Strict	95	6	198	27 sealed · 13 suppressed · 12 none · 2 ads
Balanced (default)	73	39	165	15 ads · 13 sealed · 13 suppressed · 13 none
Open	23	61	143	18 ads · 7 held · 12 suppressed · 17 none

7.3 One real ChatGPT export

The version 3 evaluator, bundled for Node together with a small driver (SHA-256 16e906a5...), was run on the owner's computer against the unzipped export and printed only counts. Each chat was evaluated as one turn: the concatenation of the user's own messages (assistant messages excluded, at most 200,000 characters), with no Episodic history and no ads yet shown. The profile interests were set to synthetic values (rust, trail running, therapy), not the owner's, so that the Semantic tier and a sensitive interest were exercised. The run took about 12 seconds.

Measure	Value
Conversation files; chats; non-empty	7; 602; 601
User messages	9,295
Chats with at least one category label	369
Chats flagged sensitive (INV-3)	40: politics 25, health 5, religion 5, financial hardship 4, mental health 1
Chats with identity cues, by kind	email 25, age 21, phone 6, location 6, personal name 4, occupation or income 2
Invariant breaches, any preset, any world	0

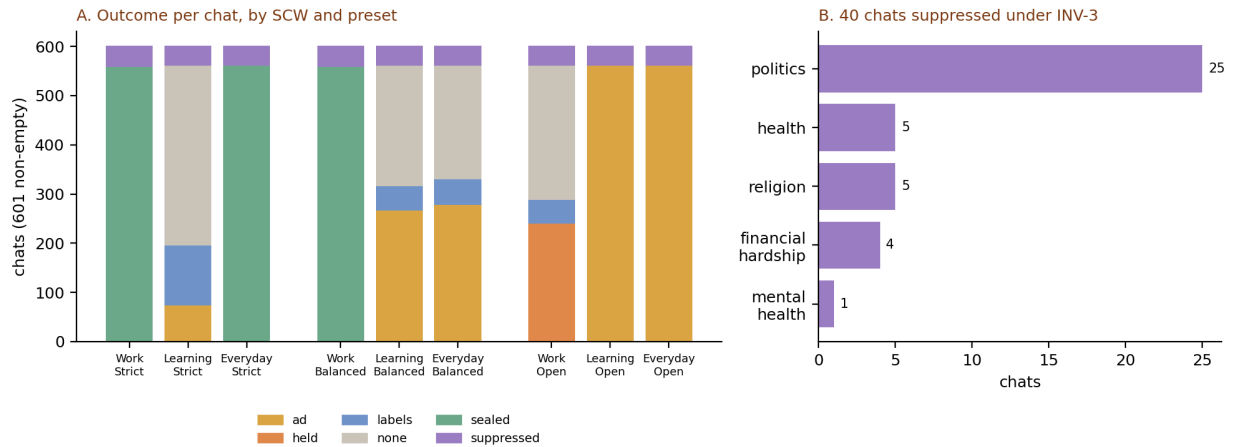


Figure 6. Measured on one real export. A: outcome per chat for each world under each preset. Work’s suppressed count is 43 rather than 40 because its blocked topics add three chats. B: the 40 chats suppressed under INV-3, by the first sensitive class matched.

Preset	Work	Learning	Everyday
Strict (95)	sealed 558 · suppressed 43 · 0 released	ad 74 · labels 121 · none 366 · suppressed 40 · 467 released	sealed 561 · suppressed 40 · 0 released
Balanced (73)	sealed 558 · suppressed 43 · 0 released	ad 267 · labels 49 · none 245 · suppressed 40 · 1,521 released	ad 278 · labels 52 · none 231 · suppressed 40 · 1,643 released
Open (23)	held 240 · labels 48 · none 273 · suppressed 40 · 1,332 released	ad 561 · suppressed 40 · 2,073 released	ad 561 · suppressed 40 · 3,034 released

Four findings follow. First, the invariants held on real data: across nine world-and-preset runs, no released label was from a sealed tier, identity-shaped or sensitive, no sensitive chat released anything, and the sensitive interest therapy was never released. Second, the presets differ in kind, not just degree. Strict lets only the Learning world read the current message, and only for education ads: 74 of 601 chats would carry an ad. Balanced opens two worlds to contextual ads and roughly 45 per cent of chats would carry one. Open lets every non-sensitive chat carry an ad in two worlds and holds 240 for approval in Work. Third, suppression is the same across presets (40 chats), because INV-3 does not depend on any setting. Fourth, the same person’s history yields 0, 1,643 or 3,034 released labels in the Everyday world depending on the policy alone, which is the quantity the global switches in ChatGPT cannot express.

7.4 How much the classifier can be trusted

There is no labeled ground truth for this export, so precision and recall remain unmeasured. The classifier is the one introduced in version 2.3, where the rule behind each flag was counted on the same export: a looser phone pattern had matched version strings and IDs (60 chats, of which 51 had no other identity cue), and the word campaign had inflated political flags. Version 2.3 tightened the phone pattern to grouped formats, removed campaign, and neutralized compounds such as technical debt before sensitive matching. Version 3 keeps those changes; the counts above (6 phone matches, 25 political chats) reflect them.

Fewer flags is not the same as better flags. A missed identity cue does not leak text, because the evaluator never releases text; a missed sensitive topic does matter, because the turn would not be suppressed. That is why the sensitive vocabulary was narrowed only where a commercial or technical reading dominates, and why a learned classifier behind the same evaluator remains the clearest next step.

8 Threat model and limitations

Limit	Consequence	Mitigation
No enforcement inside ChatGPT	The pasted profile is a request; the ad selector does not read it	The page points to OpenAI's enforced controls and marks the instruction as advisory
Model of the selector, not the selector	The payload and replays show what the policy permits, not what OpenAI reads; OpenAI also uses general location and language [1]	Stated on the page and in this paper
Keyword classifier	Misses paraphrase; can misfire on words out of context	Conservative defaults; precision fixes with tests; errors push toward withholding
Per-chat replay	Each chat was one turn with no history; multi-turn Episodic effects on real data are unmeasured	Episodic behavior is covered by the property test on synthetic histories
One export	Schema and findings may not generalize; the export format may change	Findings are stated as observations about one export
Scope of version 3	No in-page export audit, memory editor or device-stored profile (all present in version 2.3); search history and the daily cap are advisory only	Stated in the README and on the Apply page
Evaluator coverage	A logical boundary fails if any egress path bypasses the evaluator	In the page the evaluator is the only producer of released labels; elsewhere this must be audited
Viewer	Rendering inside the claude.ai artifact viewer is not yet tested	Headless Chromium at two widths; the file uses no viewer-specific features

The first row matters most. The application gives a user an exact account of their own policy and a correct list of the switches that matter, but it cannot make a provider honor a policy the provider has not adopted. A privacy tool that overstates its force does harm, because users stop looking for the controls that do work.

9 Discussion and conclusion

The logical SCW is useful wherever a physical boundary is unavailable: inside another vendor's product, between tenants sharing a context store, or between agents sharing one model context. The ingredients are always the same four: a declarative object, one evaluator on every egress path, invariants that dominate configuration, and a reason for every withholding. Typed memory adds a fifth: when tiers are kinds rather than durations, whole kinds (identity, instructions) can be removed from an egress path by construction instead of by a setting.

The export points to a concrete provider-side change. The ad system’s inferred interests are already a structured Semantic object that the user can download. A provider that let users review those interests one at a time, and grant them per kind of conversation, would give Free users a control between “personalize everything” and “clear everything”. Where a provider controls the runtime, the same evaluator could sit in front of the ad selector, and the application’s payload would become the selector’s real input rather than a model of it.

Publishing the application openly is intentional. Its value is in its rules, not in keeping them secret; a reader who reverse engineers the page learns exactly what this paper states. Natural extensions, none built, are a learned classifier behind the same evaluator, a formal proof of invariant dominance, an executed promotion gate for user-approved memory, and connecting logical SCWs to the runtime SCW gates of Maxey0-SuperSpace so that one policy object governs both.

A context boundary does not need to be a process to be exact.

Written as an object, enforced by one pure function, dominated by a few invariants, typed by memory kind and required to explain itself, a boundary can be computed, tested on a person’s own data and shown to the person it protects. The tests reported here support the invariants on the inputs tested. What remains unmeasured, and what a provider would still have to adopt, is stated as such.

Author statement

Mike Cohan (Human0) conceived the SCW Privacy project, the Structured Context World architecture, the Maxey0 memory model and the purpose of the application; directed the work; and supplied his own ChatGPT data export as the dataset, on the condition that none of it be built into the application. The first version of the application was built by the Maxey1 agent (Claude Sonnet 4.6). Maxey0-Claude, the orchestrating Claude agent of the SCW0 project (session configured as Claude Opus 5.5), built versions 2 to 3, designed the evaluator and its rule order, wrote and ran the tests and the export replays, verified OpenAI’s documented controls, and drafted this paper with Mike Cohan. Maxey0-Claude read the export’s schema, aggregate counts and two boolean account flags only, never its message text, interest text, email, birth year or account ID. Contact: human@maxey0.com.

Note on sources and claims

Statements about OpenAI’s ads, Temporary Chat and data export are taken from OpenAI’s Help Center [1][2][3], read on 29 September 2026, and may change. Statements about the export describe one export dated 11 September 2026. Statements about Maxey0, SCWs, SCW0 and the memory model represent the research program [5][6]. Every number in Section 7 was produced by executing the code named in Appendix B. Figure 6 plots those measured counts; Figures 4 and 5 are screenshots of the application. The advertisers in the application are fictional, and the payload shown in the Advertiser Lens is the application’s illustration, not OpenAI’s format.

References

- [1] OpenAI Help Center. “Ads in ChatGPT.” <https://help.openai.com/en/articles/20001047-ads-in-chatgpt> (accessed 29 September 2026).
- [2] OpenAI Help Center. “Temporary Chat FAQ.” <https://help.openai.com/en/articles/8914046-temporary-chat-faq> (accessed 29 September 2026).

- [3] OpenAI Help Center. “How do I export my ChatGPT history and data?” <https://help.openai.com/en/articles/7260999-how-do-i-export-my-chatgpt-history-and-data> (accessed 29 September 2026).
- [4] Cohan, M., and Maxey0-Claude. Maxey0 SCWs for ChatGPT Ads, version 3 (Maxey0-SCWs_For_ChatGPT_Ads.tsx). Maxey0, 30 September 2026.
- [5] Maxey0-ChatGPT 5.5 and Cohan, M. Structured Context Worlds: architectural controls for safe autonomous agents. Maxey0, 2026.
- [6] Cohan, M., and Maxey0-ChatGPT 5.5. Maxey0 System One: a provider-neutral decision layer for Structured Context Worlds. Maxey0, 2026.
- [7] Prior work in information-flow control, capability security, data minimization and privacy-by-design supplies the underlying mechanisms. The contribution here is their composition as a user-side logical SCW over typed memory, not the invention of those mechanisms.

Appendix A: export schema observed

Path	Type	Tier
ads.json > ads_profile[] > profile_version	int	Semantic
ads.json > ads_profile[] > inferred_profile > user_interests[] > interest	string	Semantic
ads.json > ad_history[] > event_name, event_time, advertiser_name, ad_title	strings	Episodic
ads.json > ad_hides[], reported_ads[]	lists	Episodic
user.json > email, birth_year, chatgpt_plus_user, id	str, int, bool, str	Persistent
user_settings.json > [0] > settings > training_allowed	bool	Persistent
conversations-NNN.json > [] > memory_scope, is_do_not_remember, create_time	str, bool, float	Episodic
conversations-NNN.json > [] > mapping > * > message > author.role, content.parts[]	str, list	Episodic (user text → labels)

Appendix B: reproducing the results

Application source: Maxey0-SCWs_For_ChatGPT_Ads.tsx (SHA-256 ad921ad5...); the classifier, evaluator, import, export, score and presets are named exports. The tests and the export replay were run on build 4e12a49d..., which differs from it only in display text (“window” renamed to “world”); the engine code is identical. Property test: prop.ts, bundled with esbuild and run with Node; a linear congruential generator with seed 0x5c3a makes its counts reproduce exactly, and setting N and a mutated copy of the source reproduces the mutation check. Real export: audit.ts, bundled with the application to one Node file (SHA-256 16e906a5...) and run as node audit.cjs <export folder>; it prints aggregate counts only. Browser checks: the application bundled with React and driven by Playwright in headless Chromium at 1440 and 400 pixels. This does not exercise the claude.ai artifact viewer itself.